

PairPath PP2 Evidence

Everything needed for the PP2 slides in one place: architecture, technologies, component flow, model evidence, and a requirement-by-requirement assessment. Every number here was measured from the running system, not estimated.

1 • Architecture

THREE SERVICES, ONE RESPONSIBILITY EACH

SERVICE	PORT	RESPONSIBILITY
frontend	3000	Shared editor, chat, role controls, intervention display
api	3001	Auth, sessions, event logging, code execution, real-time gateway
ml-service	8000	Feature extraction, state classification, intervention choice, hint retrieval

The browser never calls the ML service directly — the API gateway coordinates everything, so authentication and session context are enforced in one place.

DATA STORES

- **PostgreSQL** — users, sessions, roles, every behavioural event, interventions, peer reviews. The research record.
- **MongoDB** — an analytics trail of every prediction with the features it used.
- **Redis** — intervention cooldowns. Falls back to in-memory if unavailable.

2 • Technologies

LAYER	TECHNOLOGY	WHY
Frontend	Next.js 14, React, TailwindCSS	Server-rendered pages, fast iteration
Editor	Monaco	Same engine as VS Code; supports read-only enforcement
Real-time	Socket.IO	Persistent two-way connection for live collaboration
Backend	NestJS (Node.js, TypeScript)	Modular structure, dependency injection, typed end to end
ORM	Prisma	Type-safe database access, schema as source of truth
ML service	FastAPI (Python)	Fast, typed request/response contracts via Pydantic
Classifier	XGBoost	Strong on small tabular data; explainable feature importance
Execution	Docker (JDK 17)	Isolated sandbox for running student Java
Auth	JWT (Passport)	Same token secures REST and the socket handshake

3 • How my component works

THE LOOP, IN SIX STEPS

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|------------------------------|--|
| 1. TWO STUDENTS PAIR | One drives (types), one navigates (guides).
The navigator's editor is read-only – roles are enforced by the system, not by agreement. |
| 2. EVERY ACTION LOGGED | Edits, code runs and their results, chat messages, role swaps – all written to the database as they happen. |
| 3. THE MODEL IS ASKED | Every 60 seconds, every 30 actions, and instantly after a failed run. It reads the last 3 minutes and computes 15 behavioural measures. |
| 4. A STATE IS PREDICTED | One of five: productive, driver dominance, passive navigator, logic struggle, disengaged. |
| 5. AN INTERVENTION IS CHOSEN | Only if confidence is high enough, only if the same nudge has not fired in 5 minutes, and addressed to the pair or to one student. |
| 6. THE STUDENTS SEE IT | A glow, a pulse, a highlight, or a hint card. Accepting or dismissing it is recorded. |

The design rule behind all of it: a message says *where to look* and *what to do differently* — never how to fix the code. That is enforced by the shape of the message contract, so a solution cannot be delivered even by accident.

4 • Model, dataset and hint pipeline

THE MODEL

- **XGBoost**, 5-class classifier
- **15 behavioural features** over a 180-second sliding window
- Hyperparameters selected by **grouped 5-fold cross-validation** across 108 configurations — not hand-picked
- Every trained model writes a **model card**: dataset, split, held-out sessions, metrics

THE DATASET

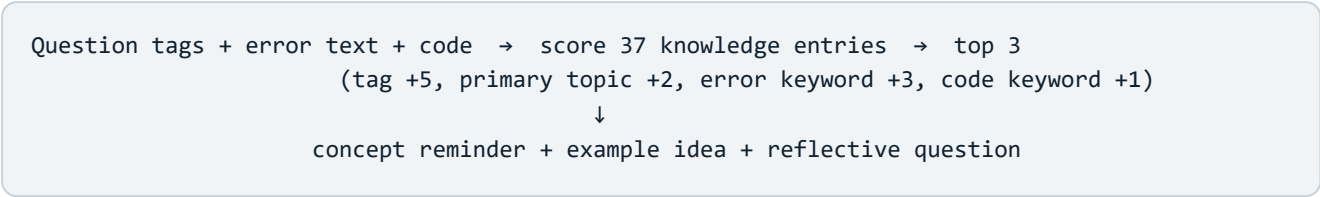
- **200 sessions** → **3,709 labelled windows**
- Split **by whole session**, never by row, so no session appears on both sides
- Duplicates removed within sessions; class imbalance handled by **weighting**, not by copying rows
- Trainer **refuses to run** on labels that are not human-sourced

THE 15 FEATURES

GROUP	MEASURES
Activity	total_edit_count, driver_edit_count, navigator_edit_count, edit_balance_r
Execution	run_attempt_count, run_success_rate, consecutive_failure_count, error_rec
Communication	discussion_note_count, navigator_note_count
Roles & time	role_switch_count, seconds_since_role_switch, session_elapsed_seconds

Two features carry the hardest distinctions. *Navigator message count* separates driver dominance from a passive navigator — both look like "one person typing"; the difference is whether the other student is talking. *Seconds since role switch* separates dominance from a healthy pair.

THE HINT PIPELINE (RAG-LITE)



Retrieval runs over a **curated corpus** rather than a generative model. That is a deliberate safety guarantee: **no document in the corpus contains the solution to an exercise**, so a hint cannot leak an answer. If nothing matches, a safe general hint is returned rather than an error.

RETRIEVAL MEASURED OVER 26 LABELLED CASES	ACCURACY	UNWANTED FALLBACKS
Concept tags only	69.2%	6 of 24
Tags + error + code	100%	0 of 24

Supplying the error message is worth +30.8% retrieval accuracy. Without it, a pair hitting an array error on a question tagged *modulo* receives generic advice; with it, they receive the array-bounds concept. Reproducible via `dev_tools/evaluate_rag.py`.

The 100% is over cases authored while building the system — it shows every situation anticipated is handled, not that retrieval is perfect. Real student errors would be expected to lower it.

5 • Model evidence

HEADLINE NUMBERS

0.891 overall accuracy on held-out sessions	0.871 macro F1 average across all 5 states	+0.283 improvement over a rule-based baseline	1.3 ms mean response time p99 under 3 ms
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COLLABORATION STATE	PRECISION	RECALL	F1	WINDOWS
Disengaged	1.000	1.000	1.000	117
Passive navigator	0.897	0.991	0.942	114
Logic struggle	0.971	0.868	0.917	228
Driver dominance	0.741	0.901	0.813	152
Productive	0.821	0.692	0.751	133
Macro average	0.886	0.890	0.884	744

WHAT PRECISION AND RECALL MEAN — IN PLAIN WORDS

Precision — when the system says a state, how often is it right?
Passive navigator 0.897 → about 9 out of 10 times it flags a silent navigator, it is correct.

Recall — of all the times a state really occurred, how many did it catch?
Passive navigator 0.991 → it almost never misses one.

F1 — the two combined into a single score. Useful because a system can cheat on one by sacrificing the other.

Where they trade off: driver dominance has high recall (0.901) but lower precision (0.741) — it catches nearly every case, at the cost of some false alarms. Given the intervention is a gentle "consider switching roles," that is the safer direction to err in.

METRICS THAT DO NOT APPLY, AND ONE NOT YET MEASURED

METRIC	STATUS	REASON
Mean squared error	N/A	MSE measures error in <i>predicted numbers</i> . This is classification — the output is a category, not a value. Precision, recall and F1 are the correct measures.
Confidence threshold	IMPLEMENTED	The system stays silent below 0.6 confidence. Calibration against annotated data is planned.
Usability score (SUS)	NOT MEASURED	No usability study has been run. Design decisions support it, but there is no score — say this rather than estimate one.

6 • Functional requirements

FR	STATUS	EVIDENCE / GAP
FR1 Input handling	PARTIAL	Captures code, typing activity, code runs and results, discussion, role switches, peer review submissions — seven event types. Gap: inputs from the syntax analyser and feedback generator are not consumed; those are sibling components not yet integrated.
FR2 Data analysis	PARTIAL	Identifies collaboration patterns (5 states) and detects communication problems (passive navigator). Gap: peer feedback <i>quality</i> is not evaluated — the review captures scored answers, not an assessment of feedback quality.
FR3 Core functionality	SATISFIED	Real-time monitoring with three triggers. Unbalanced participation → driver dominance. Weak communication → passive navigator. Both produce support.
FR4 Output & feedback	MOSTLY	Prompts for participation, communication, and reasoning (reflective questions) all delivered in real time. Gap: no prompt targets peer review quality.
FR5 User support	SATISFIED	Every hint is a concept reminder, an example idea and a reflective question — designed to prompt explanation and discussion rather than supply answers.
FR6 Output presentation	PARTIAL	On-screen messages, toasts, and highlighted interface panels all working. Gap: highlights are on interface panels, not in-editor code lines; peer review is question-based rather than line-based comments.
FR7 System integration	NOT YET	Contracts are documented and the API is JSON/REST ready, but no live integration with the syntax analyser, feedback generator or study guider exists.

7 • Non-functional requirements

NFR	STATUS	EVIDENCE
1 • Performance	SATISFIED	Target <100 ms. Measured 1.3 ms mean, 3.0 ms at the 99th percentile — 33× inside budget.
2 • Reliability	SATISFIED	Deterministic: identical input always gives identical output. Evaluation reproduces to the same figures. Every external dependency has a fallback — the system degrades rather than fails.
3 • Usability	NOT MEASURED	Designed for it — non-intrusive nudges, self-dismissing praise, no jargon, sensitive prompts sent privately. But no usability study has been conducted, so there is no score.
4 • Efficiency	SATISFIED	15 features, 630 KB model, ~1 ms inference. Cooldowns cap intervention frequency; the model is queried on triggers rather than continuously.
5 • Security & privacy	MOSTLY	JWT verified on both REST and socket handshake; identity taken from the token, never the message; session membership enforced on every action; student code runs in an isolated container with no network. Outstanding: ethical clearance before real data collection.
6 • Interoperability	READY	REST + JSON throughout, documented endpoint and event contracts, service-to-service over HTTP. Ready for integration but not yet demonstrated with sibling components.

8 • Objectives

OBJECTIVE	STATUS	EVIDENCE
1 • Structured pair programming	ACHIEVED	Driver/Navigator roles enforced <i>structurally</i> — the navigator's editor is read-only, so roles cannot quietly collapse. Either partner may take the keyboard, and rotation is monitored.
2 • Guided peer review	PARTIAL	A post-session review exists with structured questions, scoring, and tailored recommendations. Gap: not line-based, and the rubric is simple yes/no rather than explanation-focused prompts.
3 • Collaboration-aware support	ACHIEVED	The central contribution. Real-time monitoring, five-state classification, adaptive prompts targeting weak communication, unbalanced participation and ineffective interaction.
4 • Impact on learning outcomes	FUTURE WORK	Requires the classroom study. The measurement design exists — experimental vs. control with the support layer disabled — but it has not been run.
5 • Effect on collaborative skills	FUTURE WORK	Same study. Every dependent variable is derivable from data the system already records.

9 • Summary for the panel

COMPLETE

- Real-time collaborative editor with enforced roles
- Full behavioural instrumentation — seven event types
- Five-state classifier, evaluated on unseen sessions
- Confidence-gated, cooldown-limited interventions
- Role-addressed delivery for sensitive prompts
- Hint retrieval that cannot leak solutions
- Sandboxed code execution and authenticated sockets
- Post-session peer review with scoring
- Reproducible training and evaluation pipeline

REMAINING

- Integration with the three sibling components
- Peer review quality assessment
- Line-based review comments
- Usability study
- Ethical clearance
- Classroom study — objectives 4 and 5

Two things worth saying before you are asked.

1. Driver dominance and passive navigator were initially defined so similarly that no classifier could separate them. Redefining dominance as *an engaged navigator who never receives the keyboard* — which also matches the intervention each triggers — raised overall F1 substantially. That is a finding about how collaboration states should be defined, and it is arguably the most transferable result here.
2. Objectives 4 and 5 are deliberately future work. They require a classroom study, which requires ethical clearance. Naming that as a planned phase is stronger than implying it is done.

Every figure on this page was measured from the running system on 27 August 2026 — accuracy and per-class scores from the held-out evaluation, latency from 300 timed predictions.